Assignment 3

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3’rd SEM
Data Structure

Question:- Write a C program to implement a Priority Queue using three separate linear queues represented by arrays. The program should support insertion, deletion, and display operations. Elements with higher priority must be deleted before elements with lower priority.

Priority 1 – Low
Priority 2 – Medium
Priority 3 – High

Answer:-

GitHub Link :- https://github.com/bhaskargr0-spec/DS

Program :-

`priorityqueue.h`

```
#ifndef PRIORITYQUEUE_H
#define PRIORITYQUEUE_H


#include <stdio.h>


#define MAX 5


/* Three separate linear queues */

int low[MAX], medium[MAX], high[MAX];
```

```
/* Front and rear positions for each queue */
int front_low = 0, rear_low = -1;
int front_medium = 0, rear_medium = -1;
int front_high = 0, rear_high = -1;

/* Remove invalid characters from the input line */
void clear_input_buffer(void)
{
    int ch;
    while ((ch = getchar()) != '\n' && ch != EOF)
    {
        /* discard remaining input */
    }
}

/* Insert an item into the selected priority queue */
void insert(int item, int priority)
{
    if (priority == 1)
    {
        if (rear_low == MAX - 1)
            printf("Low priority queue is full.\n");
        else
        {
            low[++rear_low] = item;
            printf("%d inserted into Low Priority Queue.\n", item);
        }
    }
    else if (priority == 2)
    {
        if (rear_medium == MAX - 1)
            printf("Medium priority queue is full.\n");
        else
        {
```

```
            medium[++rear_medium] = item;
            printf("%d inserted into Medium Priority Queue.\n", item);
        }
    }
    else if (priority == 3)
    {
        if (rear_high == MAX - 1)
            printf("High priority queue is full.\n");
        else
        {
            high[++rear_high] = item;
            printf("%d inserted into High Priority Queue.\n", item);
        }
    }
    else
        printf("Invalid priority. Enter 1, 2, or 3.\n");
}

/* Delete from the highest-priority non-empty queue */
void delete_item(void)
{
    if (front_high <= rear_high)
    {
        printf("Deleted item: %d (High Priority)\n", high[front_high]);
        front_high++;
    }
    else if (front_medium <= rear_medium)
    {
        printf("Deleted item: %d (Medium Priority)\n", medium[front_medium]);
        front_medium++;
    }
    else if (front_low <= rear_low)
    {
        printf("Deleted item: %d (Low Priority)\n", low[front_low]);
```

```
        front_low++;
    }
    else
        printf("Priority Queue is empty. Deletion not possible.\n");
}

/* Display all waiting items in service order */
void display(void)
{
    int i;

    printf("\n--- Priority Queue Contents ---\n");

    printf("High Priority: ");
    if (front_high > rear_high)
        printf("Empty");
    else
        for (i = front_high; i <= rear_high; i++)
            printf("%d ", high[i]);

    printf("\nMedium Priority: ");
    if (front_medium > rear_medium)
        printf("Empty");
    else
        for (i = front_medium; i <= rear_medium; i++)
            printf("%d ", medium[i]);

    printf("\nLow Priority: ");
    if (front_low > rear_low)
        printf("Empty");
    else
        for (i = front_low; i <= rear_low; i++)
            printf("%d ", low[i]);
```

```
    printf("\n");
}


#endif
```

main.c

```
#define USE_PRIORITY_QUEUE
#include "../root.h"


int main(void)
{
    int choice, item, priority;
    int status;


    do
    {
        choice = 0;
        printf("\n===== PRIORITY QUEUE MENU =====\n");
        printf("1. Insert\n");
        printf("2. Delete\n");
        printf("3. Display\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");


        status = scanf("%d", &choice);
        clear_input_buffer();


        if (status == EOF)
        {
            printf("\nInput ended. Exiting program.\n");
            break;
        }
```

```
if (status != 1)
{
    printf("Invalid input. Please enter a number.\n");
    continue;
}

switch (choice)
{
    case 1:
        printf("Enter item: ");
        status = scanf("%d", &item);
        clear_input_buffer();

        if (status == EOF)
        {
            printf("\nInput ended. Exiting program.\n");
            return 0;
        }
        if (status != 1)
        {
            printf("Invalid item. Please enter an integer.\n");
            break;
        }

        printf("Enter priority (1 = Low, 2 = Medium, 3 = High): ");
        status = scanf("%d", &priority);
        clear_input_buffer();

        if (status == EOF)
        {
            printf("\nInput ended. Exiting program.\n");
            return 0;
        }
        if (status != 1)
```

```
                {
                    printf("Invalid priority. Please enter an integer.\n");
                    break;
                }

                insert(item, priority);
                break;

            case 2:
                delete_item();
                break;

            case 3:
                display();
                break;

            case 4:
                printf("Exiting program.\n");
                break;

            default:
                printf("Invalid choice. Please select 1 to 4.\n");
        }

    } while (choice != 4);

    return 0;
}
```

Output :-

===== PRIORITY QUEUE MENU =====

1. Insert

2. Delete
3. Display
4. Exit
   Enter your choice: 1
   Enter item: 10
   Enter priority (1 = Low, 2 = Medium, 3 = High): 1
   10 inserted into Low Priority Queue.

===== PRIORITY QUEUE MENU =====

1. Insert
2. Delete
3. Display
4. Exit
   Enter your choice: 1
   Enter item: 15
   Enter priority (1 = Low, 2 = Medium, 3 = High): 2
   15 inserted into Medium Priority Queue.

===== PRIORITY QUEUE MENU =====

1. Insert
2. Delete
3. Display
4. Exit
   Enter your choice: 1
   Enter item: 20
   Enter priority (1 = Low, 2 = Medium, 3 = High): 3
   20 inserted into High Priority Queue.

===== PRIORITY QUEUE MENU =====

1. Insert
2. Delete
3. Display
4. Exit
   Enter your choice: 3

--- Priority Queue Contents ---
High Priority: 20
Medium Priority: 15
Low Priority: 10

===== PRIORITY QUEUE MENU =====

1. Insert
2. Delete
3. Display
4. Exit
   Enter your choice: 2
   Deleted item: 20 (High Priority)

===== PRIORITY QUEUE MENU =====

1. Insert
2. Delete
3. Display
4. Exit
   Enter your choice: 2
   Deleted item: 15 (Medium Priority)

===== PRIORITY QUEUE MENU =====

1. Insert
2. Delete
3. Display
4. Exit
   Enter your choice: 2
   Deleted item: 10 (Low Priority)

===== PRIORITY QUEUE MENU =====

1. Insert
2. Delete
3. Display
4. Exit
   Enter your choice: 4
   Exiting program.